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Explainable data generation from observed small samples by matrix tri-factorization

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Data generation on small samples has exhibited great potential on real-world biomedical scenarios, especially on extreme cases such as early-stage or emerging biomedical conditions (e.g. Patient Zero of a brand-new disease with only one sample) because of insufficient sample information. Herein, we design an explainable data generation approach under extreme sample scarcity by exploiting inherent low-dimensional principles contained in observed high-dimensional data with comprehensive information, named by Explainable Data Generation (EDG) based on matrix tri-factorization. EDG aims to represent observed small high-dimensional samples by low-dimensional latent representation with a larger rather than equal sample size. This leads to explainable sample generation, due to transparency and interpretable process of decomposition and reconstruction. In experiments, EDG is tested on four tasks of classification, clustering, feature selection and tipping point prediction on simulated and real datasets (single-cell RNA-seq / ATAC-seq data) under extreme sample scarcity, with results demonstrating the superiority of EDG.

In the current data-driven era, data serves as a fundamental resource across various disciplines, including machine learning, data mining, and biomedicine. For instance, machine learning models rely heavily on large-scale datasets for training to identify patterns and produce accurate predictions. However, in scenarios with limited data availability, such as cases involving even only a single sample, model training becomes significantly challenging. This underscores the increasing demand for large-scale and high-quality datasets. Nonetheless, obtaining real-world data on a large scale is often hindered by substantial costs, poor quality, privacy constraints, and limited accessibility. To address these challenges, the techniques of data augmentation and data generation have emerged as a viable alternative, offering augmented or synthetic datasets to supplement or replace real-world data for enhancing model training.

Recently, data augmentation techniques[1–5] have been increasingly utilized to mitigate the challenges posed by insufficient model training due to limited data availability. For example, SMOTE-based (Synthetic Minority Oversampling Technique) methods[1,4,5] generate synthetic samples by linearly interpolating between existing nearest neighbors, primarily improving class balance and local diversity. Mixup-based methods[1–3] generate samples by convexly combining pairs of training examples and their labels, improving generalization and robustness. However, these methods primarily construct samples through convex combinations of existing data points, without generating entirely new data, they may struggle to capture high-dimensional dependencies present in complex biological data.

Alternatively, in scenarios where real-world data is scarce or costly to obtain, data generation techniques have emerged as a viable

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solution for producing large-scale training samples. Recently, various methods have been proposed to generate realistic synthetic data that closely mimics the characteristics of real data, including generative adversarial networks (GANs)[6–9] and variational autoencoders (VAEs)[10–13]. In detail, GANs-based models[6–9] consist of two neural networks with a generator and a discriminator that operate in a zero-sum game, enabling the generation of highly realistic synthetic data. Similarly, VAEs-based models[10–13] encode input data into a probabilistic latent space and subsequently decode it to learn the underlying data distribution, facilitating the generation of new data by sampling from this latent space. Despite their advancements, these data generation methods face inherent limitations. These deep-learning-based approaches require substantial amounts of training data and significant computational resources for effective pre-training and deployment. In cases of extremely scarce data, such as single-sample scenarios, these methods often fail to capture the intrinsic properties of real data, leading to low-quality generation. Furthermore, their black-box nature limits interpretability, reducing their credibility in practical applications. Consequently, there is an urgent need for data generation techniques that are effective, interpretable, and capable of producing high-quality synthetic data, particularly in challenging scenarios with scarce even only one sample per category availability.

Herein, to address the aforementioned challenges, we propose an explainable data generation approach, termed Explainable Data Generation (EDG), based on matrix tri-factorization. The framework of EDG is depicted in Fig. 1. This method leverages the inherent low-dimensional principles embedded within the observed high-dimensional data for new data generation, even when the sample size is limited. EDG operates under a fundamental and general assumption: the observed high-dimensional data originates from a low-dimensional latent system through unknown nonlinear or diffusive processes, yet retains comprehensive information about this latent system due to its high-dimensional features (as shown in Fig. 1a). Rather than by latent representation with an equal sample size as the observed samples in traditional methods, EDG aims to represent the observed small high-dimensional samples by low-dimensional latent representation with a larger sample size. This leads to new sample generation and further mitigating the challenges of scarce data and high-dimensional features (as illustrated in Fig. 1b). Specifically, EDG decomposes the observed high-dimensional samples into their inherent feature bases, corresponding to the low-dimensional latent system. This decomposition occurs on a larger sample size basis rather than an equal sample size. The generated feature bases not only reconstruct the original samples but also generate large numbers of new samples with rich diversity and creditability, which is explainable due to a linear combination process of sample reconstruction. In EDG, matrix tri-factorization cooperating with Hilbert-Schmidt Independence Criterion (HSIC)[14,15] and Optimal Transport (OT)[16,17] is utilized to guarantee the generation process stable with the explicit guidance of keeping consistent with the global structure and distribution of the original data for downstream predictive tasks. In detail, HSIC is employed to maximize the distributional consistency between the generated low-

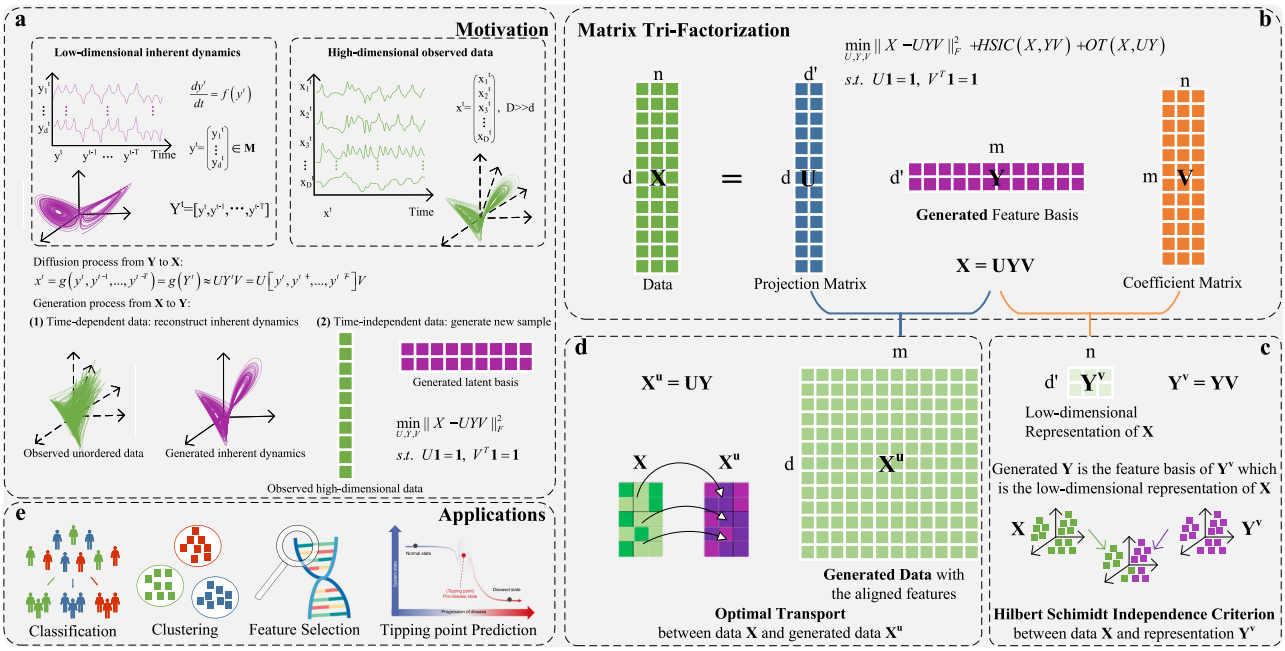


Fig. 1 | The framework of Explainable Data Generation (EDG) based on matrix tri-factorization. a The motivation of the proposed EDG. It is based on a fundamental and general assumption: high-dimensional observed data originates from a low-dimensional latent system through unknown nonlinear or diffusive processes, with the comprehensive information about the underlying latent system due to the high-dimensional features. Specifically, this work mainly focuses on the application of EDG on time-independent data for explainable data generation. **b** The main framework of the matrix tri-factorization. EDG decomposes the observed high-dimensional samples $\boldsymbol{X}$ to their inherent feature bases $\boldsymbol{Y}$, corresponding to the low-dimensional latent system, in not an equal sample size (n) but a larger sample size (m) basis. Through this process, such generated feature bases $\boldsymbol{Y}$ not only reconstruct the original samples $\boldsymbol{X}$ with HSIC (as shown in (**c**)) but also generate large numbers of new samples $\boldsymbol{X}^u$ with OT (as shown in (**d**)) with aligned features, which is explainable due to a linear combination process of sample reconstruction. In EDG, matrix tri-factorization cooperating with HSIC and OT is utilized to guarantee the generation process stable with the explicit guidance of keeping consistent with the global structure and distribution of the original data while reducing overfitting. In detail, (c) HSIC is employed to maximize the distributional consistency between the generated low-dimensional representation $Y^v = YV$ and the original data X, to ensure that the low-dimensional representation accurately captures the structure of the observed data. **d** OT is employed to ensure the alignment and consistency between the distributions of the original high-dimensional data X and the newly generated data $X^u = UY$ so that the generated samples can accurately reflect the original data's structure and distribution. **e** The proposed EDG is a general data generation framework applicable to a wide range of real-world applications, such as classification, clustering, feature selection and tipping point prediction, particularly for datasets with limited sample size.

dimensional representation and the original data, to ensure that the low-dimensional representation accurately captures the structure of the observed data (as shown in Fig. 1c). OT is employed to ensure the alignment and consistency between the distributions of the original high-dimensional data and the newly generated data so that the generated samples can accurately reflect the original data's structure and distribution, with reinforcing interpretability of the proposed EDG (as shown in Fig. 1d).

Furthermore, the explainable nature of EDG is a significant advantage over black-box deep-learning approaches. The matrix tri-factorization provides a clear and interpretable decomposition and reconstruction, enhancing the transparency of the data generation process. Besides, HSIC and OT is cooperated with matrix tri-factorization with explicit guidance of keeping consistent with the global structure and distribution of the original data. Moreover, the generated samples serve as feature bases for the original samples, enabling the generated data to be both explainable and biologically meaningful. Overall, these characteristics not only improve trust in the generated data but also enable researchers to gain deeper insights into the underlying structure of the data, enabling the generated data explainable and faithful preservation of biological characteristics. This is crucial in fields like healthcare, genomics, biomedical and finance.

The proposed EDG is a general data generation framework applicable to a wide range of real-world applications, particularly for datasets with extremely limited sample size, such as early-stage or emerging biomedical conditions (e.g., Patient Zero of a brand-new disease with only one sample). To validate its effectiveness, extensive experiments were conducted on four key tasks of classification, feature selection, clustering and tipping point prediction, on both simulated and real-world datasets (single cell RNA-seq / ATAC-seq data) with scarce samples, including extreme cases with only a single sample per category. The results consistently demonstrated the superiority of EDG in enhancing classifier performance, improving the robust of clustering centroid, selecting discriminative features and accurately predicting tipping points.

In summary, in this work, we propose an explainable data generation approach under extreme sample scarcity by exploiting inherent low-dimensional principles contained in the observed high-dimensional data, named by Explainable Data Generation (EDG), based on matrix tri-factorization. Different from modern generation methods requiring a certain amount of samples for pre-training with limited interpretability, reducing their credibility in practically biomedical applications, EDG aims to exploit high-dimensional information rather than relying on large numbers of training samples for new data generation. Furthermore, rather than by latent representation with an equal sample size as the observed samples in traditional methods, EDG represents the observed small high-dimensional samples by a low-dimensional latent representation with a larger sample size. This leads to explainable sample generation, due to transparency and an interpretable process of decomposition and reconstruction in matrix tri-factorization cooperating with OT and HSIC. In experiments, EDG is tested on four tasks of classification, clustering, feature selection and tipping point prediction on simulated and real datasets (single-cell RNA-seq / ATAC-seq data) under extreme sample scarcity, with results demonstrating the superiority of EDG.

Results

Motivation

In this section, we firstly present a comprehensive explanation of the proposed methods, which are designed as general frameworks for data generation for time-independent and time-dependent data. The workflow is illustrated in Fig. 1.

Our method follows a fundamental and general assumption that the observed high-dimensional data originates from a low-dimensional latent system via unknown nonlinear dynamical processes, but contain the comprehensive information of such a latent system due to its diffusively projected high-dimensional features (as shown in Fig. 1a). Roughly speaking, letting the latent low-dimensional dynamics, which generates the observed high-dimensional data, as $\frac{dy^t}{dt} = f(y^t)$, where $y^t = (y_1^t, y_2^t, \ldots, y_d^t)$, this dynamics is assumed to diffusively generate the observed high-dimensional data $x^t = (x_1^t, x_2^t, \ldots, x_D^t)$ with $d << D$. Consequently, the diffusively dynamical process from the low-dimensional latent representation $\boldsymbol{Y}$ to the observed high-dimensional data $\boldsymbol{X}$ is described as follows:

$$\begin{aligned} x^t &= g(y^t, y^{t-1}, \ldots, y^{t-T}) = g(\boldsymbol{Y}^t) \\ &\approx \boldsymbol{U}\boldsymbol{Y}^t\boldsymbol{V} = \boldsymbol{U}\left[y^t, y^{t-1}, \ldots, y^{t-T}\right]\boldsymbol{V} \\ &= \boldsymbol{X}^{\boldsymbol{u}t}\boldsymbol{V} = \left[x^{ut}, x^{ut-1}, \ldots, x^{ut-T}\right]\boldsymbol{V} \end{aligned} \tag{1}$$

where $\boldsymbol{Y}^t = (y^t, y^{t-1}, \ldots, y^{t-T})$ are the reconstructed samples in the latent space, and $\boldsymbol{X}^{\boldsymbol{u}t} = \left(x^{ut}, x^{ut-1}, \ldots, x^{ut-T}\right)$ are the generated samples in the observed space. U, V are normalized matrices with $\hat{U}\mathbf{1} = \mathbf{1}$, $\hat{V}^T\mathbf{1} = \mathbf{1}$. $\boldsymbol{U}\boldsymbol{Y}^t\boldsymbol{V}$ represents the linearization of the function $g(\cdot)$. Note that for traditional dimension reduction of this one sample case, $x^t = \boldsymbol{U}\boldsymbol{Y}^t\boldsymbol{V} = \boldsymbol{U}\boldsymbol{Y}^{\boldsymbol{v}t} = \boldsymbol{U}\left[y^{ut}\right]$.

This representation implies that from only one observed sample in the observed space x, we can construct multiple latent samples by $\boldsymbol{Y}^t$ in the latent space y, i.e., T samples, which actually can further generate new T samples by $\boldsymbol{U}\boldsymbol{Y}^t = \boldsymbol{X}^{\boldsymbol{u}t}$ in the observed space x. Actually, such a representation is also consistent with the spatiotemporal information transformation (SIT) framework[18–21] which is for the time-dependent data.

Clearly, with multiple observed samples for both time-dependent data or time-independent data, we can have the following representation.

$$\boldsymbol{X} = \boldsymbol{U}\boldsymbol{Y}\boldsymbol{V} = \boldsymbol{X}^{\boldsymbol{u}}\boldsymbol{V} = \boldsymbol{U}\boldsymbol{Y}^{\boldsymbol{v}} \tag{2}$$

where $\boldsymbol{X}$ is the matrix of the observed n samples with dimension d, and $\boldsymbol{Y}$ is the matrix of the reconstructed m latent samples with the dimension d' which generates the corresponding new m samples $\boldsymbol{X}^{\boldsymbol{u}} = \boldsymbol{U}\boldsymbol{Y}$ in the original space. Note that $m > n$, $d > d'$, and $\boldsymbol{Y}^{\boldsymbol{v}} = \boldsymbol{Y}\boldsymbol{V}$ is the n samples with the reduced dimension d' corresponding to the original samples $\boldsymbol{X}$, which is the traditional dimension reduction of $\boldsymbol{X}$.

Explainable data generation (EDG) based on matrix tri-factorization

Herein, we mainly focus on explainable data generation for time-independent data, as shown in Fig. 1a(2), with its main workflow illustrated in Fig. 1b, by employing matrix tri-factorization. Mathematically, consider an observed data matrix $\boldsymbol{X} \in \mathbb{R}^{d\times n}$ consisting of n samples, each characterized by d-dimensional features. Unlike traditional methods using low-dimensional latent representation with an equal sample size as the observed samples, the proposed EDG aims to represent the observed small (n) high-dimensional (d) samples X by low-dimensional (d') latent representation $\boldsymbol{Y}$ with a larger sample size (m) via matrix tri-factorization, thus leading to new sample generation and also mitigating the challenges of scarce data and high-dimensional features. In detail, EDG decomposes the observed high-dimensional samples $\boldsymbol{X}$ to their intrinsic feature bases $\boldsymbol{Y}$, corresponding to a low-dimensional latent system, in not an equal sample size but a larger sample size basis, as indicated in Eq.(2). Thus, such generated feature bases $\boldsymbol{Y}$ not only reconstruct the original samples $\boldsymbol{X}$ (as shown in Fig. 1c) with HSIC[14,15], but also generate large numbers of new samples $\boldsymbol{X}^{\boldsymbol{u}}$ (as shown in Fig. 1d) with OT[16,17], with rich diversity and aligned features. This process ensures explainability through the linear combination of sample reconstruction, further enhancing the interpretability of the generated data. The process of matrix tri-factorization for new sample

generation is defined as

$$\min_{\boldsymbol{U},\boldsymbol{Y},\boldsymbol{V}} ||\boldsymbol{X}-\boldsymbol{UYV}||_F^2 \quad s.t.\boldsymbol{U}\mathbf{1}=\mathbf{1}, \boldsymbol{V}^T\mathbf{1}=\mathbf{1} \tag{3}$$

where $||\boldsymbol{X}-\boldsymbol{UYV}||_F^2$ represents matrix tri-factorization, aiming to decompose the observed high-dimensional data $\boldsymbol{X}$ to low-dimensional latent representation $\boldsymbol{Y}$, with $\|\cdot\|_F^2$ denoting Frobenius norm. Furthermore, $\mathbf{1}$ denotes a column vector with all elements equal to 1, which are utilized to enforce the constraints that the column sums of $\boldsymbol{U}$ and the row sums of $\boldsymbol{V}$ equal to 1. In detail, $\boldsymbol{U} \in \mathbb{R}^{d\times d'}$ serves as a projection matrix that reduces the feature dimensionality from d-dimensions to d'-dimensions, with $d' < d$. The matrix $\boldsymbol{Y} \in \mathbb{R}^{d'\times m}$ represents the generated feature basis consisting of m samples with d'-dimensional features, with $m > n$. The matrix product of $\boldsymbol{Y}^v = \boldsymbol{YV} \in \mathbb{R}^{d'\times n}$ is the low-dimensional representation of the observed data $\boldsymbol{X} \in \mathbb{R}^{d\times n}$. Consequently, Y is the feature basis of $\boldsymbol{Y}^v$ and $\boldsymbol{V} \in \mathbb{R}^{m\times n}$ is the corresponding coefficient matrix. In this way, each sample in $\boldsymbol{Y}^v$ can be reconstructed as a linear combination of the generated feature basis $\boldsymbol{Y}$ and the corresponding coefficient matrix $\boldsymbol{V}$. Furthermore, to ensure that the low-dimensional representation $\boldsymbol{Y}^v$ accurately captures the structure of the observed data, HSIC is employed to maximize the distributional consistency between the generated low-dimensional representation $\boldsymbol{Y}^v = \boldsymbol{YV}$ and original data $\boldsymbol{X}$. This allows $\boldsymbol{Y}^v$ to effectively represent $\boldsymbol{X}$ in the low-dimensional space, with $\boldsymbol{Y}$ serving as the feature basis of the original data. Finally, to generate samples in the original feature space with aligned features, the generated feature basis $\boldsymbol{Y}$ is mapped back to the original feature space of $\boldsymbol{X}$ cooperating with the project matrix $\boldsymbol{U}$. The final generated samples with aligned features is denoted as $\boldsymbol{X}^u = \boldsymbol{UY} \in \mathbb{R}^{d\times m}$, with $m > n$. In addition, OT is employed to ensure the alignment and consistency between the distributions of the original high-dimensional data $\boldsymbol{X}$ and the generated data $\boldsymbol{X}^u$ so that the generated samples can accurately reflect the original data's structure and distribution, with reinforcing interpretability and reliability of the proposed EDG.

Finally, the complete loss function of the proposed EDG framework is formulated as follows:

$$\min_{\boldsymbol{U},\boldsymbol{Y},\boldsymbol{V}} \lambda_M||\boldsymbol{X}-\boldsymbol{UYV}||_F^2 + \lambda_H HSIC(\boldsymbol{X},\boldsymbol{YV}) + \lambda_O OT(\boldsymbol{X},\boldsymbol{UY}) \quad s.t.\boldsymbol{U}\mathbf{1}=\mathbf{1}, \boldsymbol{V}^T\mathbf{1}=\mathbf{1} \tag{4}$$

where λ_M, λ_H, and λ_O are hyperparameters to balance different terms.

Furthermore, the proposed EDG is a general and versatile data generation framework applicable to various real-world applications, including classification, clustering, feature selection, and tipping point prediction, particularly for datasets with limited sample sizes, as illustrated in Fig. 1e. To evaluate the effectiveness of EDG, we conducted extensive experiments across these tasks using simulated omics and real-world single-cell datasets. The performance of EDG was quantified using widely recognized metrics, such as classification accuracy, clustering accuracy, and tipping point prediction accuracy, with detailed experiment results reported as following sections.

Experimental results of EDG on simulated omics dataset

In this section, we firstly evaluate the performance of the proposed EDG on a simulated omics dataset[22] for cancer patients characterized by a small sample size and a high-dimensional feature space, whose details are introduced in Supplementary Note 1. In our experiments, 10 samples from each category were randomly selected for new data generation and model training, while the remaining samples were reserved for testing. This experimental design ensures a rigorous evaluation of the proposed EDG framework under conditions of data scarcity and high dimensionality.

Table 1 | The experimental results of the proposed EDG on the tasks of classification, feature selection, and clustering, evaluated in terms of classification accuracy and clustering accuracy

Tasks	Origin	Generation
Classification	.889(0.003)	**.986(0.002)**
Feature Selection	.515(0.005)	**.886(0.008)**
In-sample Clustering	.600(0.000)	**.900(0.000)**
Out-of-sample Clustering	.773(0.009)	**.995(0.005)**

"Origin" represents performing the tasks on the original dataset, consisting of 10 samples per category. "Generation" represents performing the tasks on the expanded dataset, which included 50 newly generated samples by EDG per category in conjunction with the original 10 samples per category.

To evaluate the quality of the generated data in improving the training of machine learning models, we firstly assessed its impact on a classification task. The performance was quantified by classification accuracy, as reported in Table 1. The "Origin" scenario involved training a Support Vector Machine (SVM) classifier[23] on the original samples, achieving classification accuracy = 0.889, reflecting suboptimal performance due to the limited training data (10 samples) and the high feature dimensionality (3980 dimensions). In contrast, the "Generation" scenario utilized the proposed EDG to generate 50 additional samples per category, which were combined with the original 10 samples for training. This approach significantly improved classification performance, with accuracy = 0.986. The superior performance of EDG is attributed to its ability to facilitate the generation of new high-quality samples, effectively addressing the challenges of data scarcity and high-dimensional features.

To further validate the quality of the generated new data, we visualize the spatial distributions of the samples using t-distributed stochastic neighbor embedding (t-SNE)[24], as displayed in Fig. 2a. In the visualization, dots represent the original 10 samples per category, while asterisks denote the 50 newly generated samples per category produced by EDG. The results demonstrate that the generated samples exhibit a spatial distribution highly consistent with the original data, with the majority of samples well-separated among categories. These findings confirm that the proposed EDG framework can generate high-quality data with aligned and meaningful features.

For the feature selection task, we employed the variance sorting method to select the top 5% of features from the full 3980-dimensional features, separately on the original and generated samples. The quality of the selected features was then evaluated using an SVM classifier, with the performance reported in Table 1. In the "Origin" scenario, features were selected from the original 10 samples per category, with suboptimal SVM classification accuracy = 0.515. This performance reflects the challenges posed by the limited number of samples (10 per category) results in minor variance differences among features, making it difficult to identify effective features from a high-dimensional feature space (3980 dimensions). In contrast, the "Generation" scenario utilized features selected from an expanded dataset, consisting of 50 newly generated samples combined with the original 10 samples per category, with a superior accuracy = 0.886. Consequently, EDG facilitates the generation of new samples with aligned features, enhancing variance differences among features, enabling the selection of features with stronger discriminative power.

To further validate the quality and discriminative power of the selected features, we visualize their spatial distribution using t-SNE[24] (Fig. 2b), with features selected from both the original 10 samples (Fig. 2b-I,Fig. 2b-III) and generated expanded dataset (Fig. 2b-II,Fig. 2b-IV) by EDG on training (Fig. 2b-I,Fig. 2b-II) and testing groups (Fig. 2b-

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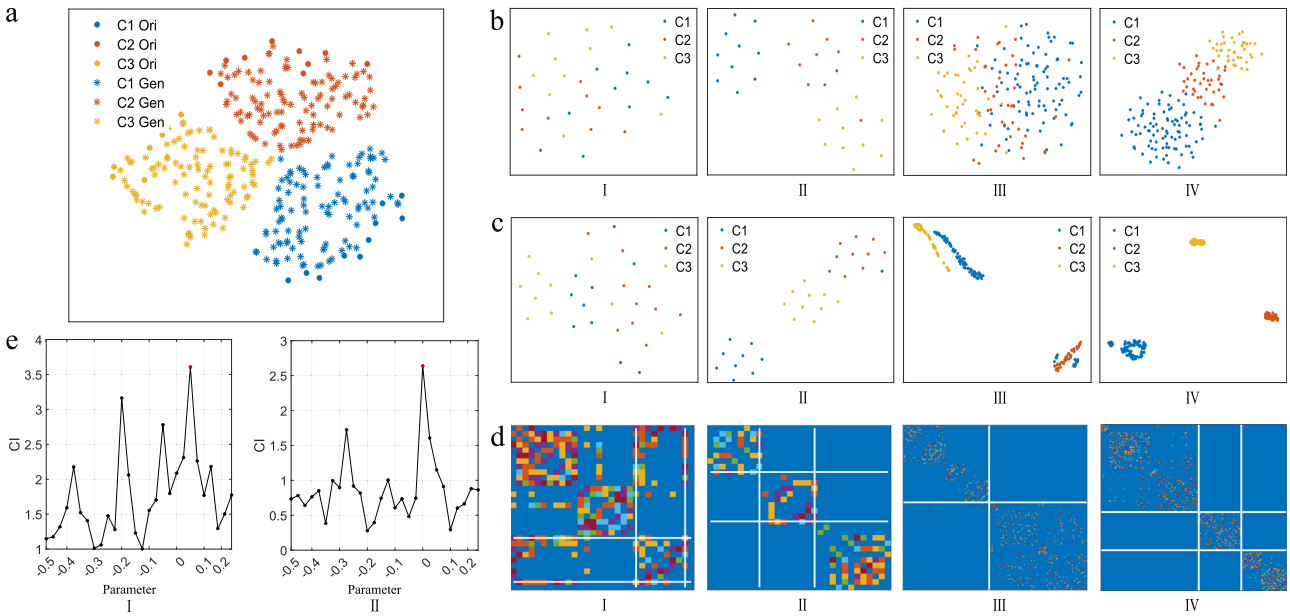


Fig. 2 | The experimental results of EDG on simulatied omics dataset. a The spatial distribution of the original and generated data produced by the proposed EDG, visualized using *t*-SNE. **b** The spatial distribution of the features selected from both the original (I, III) and generated data (II, IV) by EDG on training data (I, II) and testing data (III, IV), visualized using *t*-SNE. **c** The spatial distribution of the distances between the samples and the clustering centroids, learned from both the original (I, III) and generated data (II, IV) using EDG, for in-sample clustering (I, II) and out-of-sample clustering (III, IV), visualized using *t*-SNE. **d** The visualization of corresponding affinity matrices on the distances between the samples and the clustering centroids, learned from both the original (I, III) and generated data (II, IV) using EDG, for in-sample clustering (I, II) and out-of-sample clustering (III, IV), visualized using *t*-SNE. **e** The tipping point prediction on numerical simulation by employing dynamical network biomarkers (DNB) on both the original data (I) and generated data (II) by EDG. Source data is provided in "Source Data.xlsx".

III,Fig. 2b-IV). As observed in Fig. 2b-I and Fig. 2b-III, the samples are twisted together with unclear boundaries, and some samples belonging to different categories overlap in the same region. In contrast, Fig. 2b-II and Fig. 2b-IV exhibit a much more distinct and well-separated distribution.

For the clustering task, we employ *k*-means clustering[25] for clustering centroid learning. Specifically, we randomly select 10 samples for in-sample clustering centroid learning, while the remaining 190 samples for out-of-sample clustering evaluation by calculating the distance between the out-of-sample data and the learned centroids. The performance is quantified by clustering accuracy, reported in Table 1. In detail, 'Origin' indicates using 10 samples for clustering centroid learning, achieving unsatisfactory in-sample clustering accuracy of 0.600 and out-of-sample clustering accuracy of 0.773. The suboptimal performance is primarily due to the challenges of learning discriminative centroids from a dataset with only 10 samples, while high-dimensional (3980) features. This data sparsity and high-dimensionality hinder the accurate estimation of sample similarity, which in turn degrades the clustering performance. In contrast, in the 'Generation' scenario, clustering is performed on the generated expanded dataset. This approach yields superior performance, with in-sample clustering accuracy = 0.900 and an out-of-sample clustering accuracy = 0.995. The success is attributed to EDG enables the generated samples to effectively mitigate the challenges of scarce samples and high-dimensional features, resulting in high-quality clustering centroids with enhanced discriminative power.

To further validate the quality and discriminative power of clustering centroids, in Fig. 2c, we visualize the spatial distribution using *t*-SNE[24] of distances between samples and learned clustering centroids, learned from both the original 10 samples (Fig. 2c-I,Fig. 2c-III) and generated expanded dataset (Fig. 2c-II,Fig. 2c-IV) using EDG, for in-sample clustering (Fig. 2c-I,Fig. 2c-II) and out-of-sample clustering (Fig. 2c-III,Fig. 2c-IV). As seen in Fig. 2c-I and Fig. 2c-III, the samples are tightly clustered with unclear boundaries, and some samples from different categories overlap, both on the in-sample data (Fig. 2c-I) and out-of-sample data (Fig. 2c-III). This arises from a scarce sample size in combination with high feature dimensional, which leads to inaccurate similarity measures and hampers the learning of robust clustering centroids. In contrast, clustering centroids exhibit a much clearer and well-separated distribution on the generated expanded dataset, both on in-sample (Fig. 2c-II) and out-of-sample data (Fig. 2c-IV). Furthermore, the visualization on corresponding affinity matrices (as displayed in Fig. 2d) also revealed similar conclusions.

Finally, we evaluate the performance of the proposed EDG for tipping point prediction using a numerical simulation dataset[26–28], whose details are introduced in Supplementary Note 1. It comprises 28 time points, varying a parameter within the range of − 0.5 to 0.25, with one sample per time point. Notably, the system undergoes a critical transition at the bifurcation parameter value of 0, which corresponds to the 20th time point.

To assess the quality of generated data for assisting tipping point prediction, we employ dynamical network biomarkers (DNB)[29], with performance quantified by the sharp increase in composite index (CI) score, which is treated as the indicator of a critical state or tipping point, as shown in Fig. 2e. DNB can not work on this dataset with only one sample in each time point. Consequently, we incorporate the samples from three previous time points along with the current time point as the combined samples of the current time point for DNB and our generation approach EDG, for fair comparison. In Fig. 2e, we present the dynamic variation of CI[29] score, calculated using DNB. "Origin" condition refers to the application of DNB on the original dataset, leading to the identification of an incorrect tipping point at a parameter value other than 0. This is caused by a limited sample size, resulting in minor variance differences across time points. In comparison, the "Generation" condition refers to generate five additional samples at each time point using EDG, for tipping point prediction with DNB. This approach yields a sharp increase in CI score, peaking at a bifurcation parameter value of 0, thereby accurately predicting the

tipping point. The success of EDG can be attributed to its ability to generate high-quality samples, enhancing variance differences across time points, thereby enabling more accurate tipping point prediction.

Experimental Results on Real-World Datasets

In this section, we evaluate the quality of data generated by EDG in comparison of data augmentation methods, such as MAMix (Margin-Aware Mixup)[3], iSMOTE (Imbalance Learning via Synthetic Minority Oversampling Technique)[5], and deep generation methods, such as SCVI (Single-Cell Variational Inference)[13], cscGAN (conditional single-cell Generative Adversarial Networks)[9]. The experiments on tasks of classification, feature selection and clustering are conducted on two real-world single-cell datasets characterized by small sample size and large feature dimension. In detail, these two datasets, obtained from the studies in refs. 30,31. For simplicity, we focus on scATAC-seq dataset with 107,194-dimensional features from ref. 30 and scRNA-seq dataset with 28,930-dimensional features from refs. 30,31, respectively. Subsequently, the dataset tested on tipping point prediction is derived from prostate cancer study GSE5345 and investigates the effect of synthetic androgen R1881 on gene expression in human prostate cancer cell line LNCap[32,33]. The details of these real-word datasets are introduced in Supplementary Note 1. Herein, the following experiments cover two tasks of classification and tipping point prediction, while the other tasks of feature selection and clustering are presented in Supplementary Table 1–4 in Supplementary Note 2.

The proposed EDG is designed for real-world biomedical scenarios, especially on extreme cases such as early-stage or emerging biomedical conditions (e.g., Patient Zero of a brand-new disease with only one sample). Consequently, in the experimental settings, assuming the availability of only n samples in real-world scenarios, we randomly select n samples per category for training and data generation while using the remaining samples for testing, with n ranging from 1 to 50. Consequently, our experimental setting covers the full progression of disease development, ranging from extremely early stages with very limited sample sizes ($n \le 10$) to more data-rich regimes ($n \ge 20$). Accordingly, we report results not only in the extremely few-shot setting but also in scenarios with moderately larger sample sizes. In addition, we apply the generation methods to generate new samples in M multiples of the original number per category, combined with the original samples, with $M = 5, 10, 20$. It is important to note that deep-learning-based methods of SCVI and cscGAN typically require certain amounts of training data and computational resources for effective training and deployment. In scenarios with a limited sample size with $n \le 10$, these methods fail to generate meaningful samples. Similarly, the data augmentation methods of MAMix and iSMOTE require certain amounts of training data because they primarily construct samples through convex combinations of existing samples. They fail to generate meaningful samples in extremely limited scenarios with $n = 1, 2$. As a result, MAMix, iSMOTE, SCVI and cscGAN are unable to produce effective data in these cases, highlighting the advantages of EDG in handling scarce sample situations with even only one sample in extreme cases. Furthermore, we used multiple random splits (five repeats) to select different samples for generation, with standard deviation reported in brackets, leading to a more reliable evaluation of stability and robustness under few-shot conditions.

To assess the quality of generated data for assisting classifier training, we firstly evaluate the task of single-cell classification. The performance is quantified by classification accuracy, as reported in Table 2 and Table 3 for the datasets of scATAC-seq and scRNA-seq, respectively. In real-world scenarios with limited data availability, we assume n samples per category. 'Origin' scenario represents training an SVM classifier[23] on the original data with n samples per category. In comparison, MAMix, iSMOTE, SCVI, cscGAN, and EDG are used to generate additional samples in M multiples per category, combined with the original data for training. Specifically, on the scATAC-seq dataset in Table 2, "Origin" achieves unsatisfactory classification accuracies of 0.199, 0.325, 0.431 and 0.617, with original sample size $n = 1, 2, 5$, and 10, respectively. This suboptimal performance arises from insufficient training due to a small sample size and the high feature dimensionality. Data augmentation and deep-learning-based methods also fail to generate effective samples in these cases because their requirement of a certain amount of data for sample combination and model training, which limits their applicability for small datasets ($n \le 2$ for data augmentation and $n \le 10$ for deep learning). In comparison, EDG demonstrates superior performance, achieving classification accuracies of 0.323 ($M = 20$), 0.490 ($M = 20$), 0.584 ($M = 20$) and 0.763 ($M = 20$), with sample sizes n being 1, 2, 5, and 10, respectively. Furthermore, the "Origin" scenario yields suboptimal classification accuracies of 0.704 and 0.801, for sample sizes $n = 20$ and $n = 50$, respectively. In comparison, SCVI and cscGAN perform relatively well with accuracies being 0.725 ($M = 5$), 0.818 ($M = 5$) (SCVI) and 0.707 ($M = 5$), 0.815 ($M = 5$) (cscGAN) for sample sizes $n = 20$ and $n = 50$, respectively. Similarly, MAMix and iSMOTE achieve relatively better performance with accuracies being 0.745 ($M = 10$), 0.814 ($M = 20$) (MAMix) and 0.739 ($M = 10$), 0.813 ($M = 20$) (iSMOTE) with $n = 20$ and $n = 50$, respectively. In contrast, EDG consistently outperforms the other methods, achieving the highest accuracies of 0.798 ($M = 10$) and 0.823 ($M = 5$) for sample sizes $n = 20$ and $n = 50$, respectively. The success of EDG is attributed to exploit high-dimensional information rather than relying on large numbers of training samples for high-quality data generation consistent with the original distribution. It effectively addresses the challenges of limited data availability, especially when data is extremely scarce.

To further evaluate the robustness and generalizability of EDG, ten diverse single-cell datasets[34] spanning multiple species, organs, sequencing platforms, and biological complexities are introduced in Supplementary Table 5 in Supplementary Note 3 for the classification task. The additional benchmark includes Adam (mouse kidney)[35], Alemany-Fin (zebrafish fin)[36], Bach (mouse mammary gland)[37], Guo (human testis)[38], Karaiskos (mouse kidney)[39], Li-Mammary (mouse mammary gland)[40], MacParland (human liver)[41], Muraro (human pancreas)[42], Oihane (mouse brain)[43], and Zilionis (mouse lung)[44]. The experimental results on these ten additional datasets for the classification task is reported in Table 4. In particular, in this experimental setting, the generated multiples M is set as $M = 10$. Overall, EDG consistently achieves the best or highly competitive performance across almost all datasets and sample sizes, demonstrating its effectiveness in generating informative synthetic samples from limited observations, especially under extreme data scarcity. As sample number increases ($n = 10$ and 20), the performance gap between EDG and compared methods gradually decreases because of more samples available. Nevertheless, EDG still achieves the highest or comparable accuracy on nearly all datasets, indicating that EDG better preserves the underlying data distribution while generating diverse samples. In addition, EDG generally exhibits lower standard deviations than the original data and competing methods, indicating improved robustness and stability. Consequently, EDG provides a reliable and effective solution for generating high-quality and informative single-cell data.

Subsequently, we evaluate the performance of tipping point prediction on a real-world dataset, derived from the prostate cancer study[32,33]. Based on the experimental findings reported in ref. 32, the 2nd time point at 6 h is identified as the critical state or tipping point, signaling the onset of a pre-disease state. DNB[29] is employed with the performance quantified by a sharp increase of CI score, which is an indicator of the tipping point. The results are presented in Fig. 3. In detail, Fig. 3a illustrates the application of DNB on the original dataset with 2 samples per time point, while Fig. 3b–f indicating applying DNB on the dataset augmented with 5 additional samples per time point generated by the generation methods of MAMix (Fig. 3b), iSMOTE (Fig. 3c), SCVI (Fig. 3d), cscGAN (Fig. 3e) and EDG (Fig. 3f), respectively,

Table 2 | The experimental results of MAMix, iSMOTE, SCVI, cscGAN and the proposed EDG on the scATAC-seq dataset for the task of single-cell classification, with performance evaluated by classification accuracy

Data	Origin	Generated Sample Size in *M* Multiples														
		M=5					*M*=10					*M*=20				
		MAMix	iSMOTE	SCVI	cscGAN	EDG	MAMix	iSMOTE	SCVI	cscGAN	EDG	MAMix	iSMOTE	SCVI	cscGAN	EDG
n=1	.199(.105)	–	–	–	–	.258(.107)	–	–	–	–	.303(.086)	–	–	–	–	**.323(.075)**
n=2	.325(.128)	–	–	–	–	.463(.109)	–	–	–	–	.475(.073)	–	–	–	–	**.490(.070)**
n=5	.431(.080)	.465(.105)	.459(.095)	–	–	.553(.085)	.479(.108)	.463(.096)	–	–	.577(.079)	.482(.096)	.475(.086)	–	–	**.584(.068)**
n=10	.617(.079)	.668(.064)	.657(.070)	–	–	.727(.065)	.673(.055)	.668(.058)	–	–	.753(.045)	.695(.045)	.686(.045)	–	–	**.763(.032)**
n=20	.704(.051)	.735(.045)	.730(.048)	.725(.045)	.707(.059)	.769(.062)	.745(.045)	.739(.047)	.720(.040)	.711(.041)	**.798(.050)**	.738(.033)	.729(.038)	.710(.030)	.705(.031)	.787(.038)
n=50	.801(.009)	.810(.005)	.809(.009)	.818(.008)	.815(.009)	**.823(.005)**	.807(.007)	.805(.005)	.816(.006)	.811(.007)	.820(.005)	.814(.002)	.813(.004)	.809(.003)	.803(.006)	.819(.006)

Table 3 | The experimental results of MAMix, iSMOTE, SCVI, cscGAN and the proposed EDG on the scRNA-seq dataset for the task of single-cell classification, with performance evaluated by classification accuracy

Data	Origin	Generated Sample Size in *M* Multiples														
		M=5					*M*=10					*M*=20				
		MAMix	iSMOTE	SCVI	cscGAN	EDG	MAMix	iSMOTE	SCVI	cscGAN	EDG	MAMix	iSMOTE	SCVI	cscGAN	EDG
n=1	.181(.070)	–	–	–	–	.248(.072)	–	–	–	–	.258(.047)	–	–	–	–	**.266(.057)**
n=2	.274(.046)	–	–	–	–	.341(.040)	–	–	–	–	**.347(.049)**	–	–	–	–	.340(.042)
n=5	.424(.041)	.452(.043)	.445(.037)	–	–	.481(.033)	.459(.038)	.455(.033)	–	–	.487(.030)	.461(.030)	.459(.035)	–	–	**.492(.030)**
n=10	.432(.039)	.505(.035)	.499(.039)	–	–	.532(.030)	.507(.038)	.504(.035)	–	–	.531(.028)	.512(.032)	.509(.036)	–	–	**.538(.029)**
n=20	.618(.025)	.628(.025)	.625(.024)	.620(.022)	.618(.020)	**.637(.023)**	.629(.023)	.627(.026)	.622(.022)	.629(.027)	.635(.024)	.630(.020)	629.(.025)	.626(.026)	.629(.019)	.634(.022)
n=50	.717(.008)	.719(.003)	.719(.008)	.717(.007)	.720(.003)	**.721(.005)**	.719(.007)	.719(.008)	.718(.009)	.715(.007)	.719(.005)	.716(.005)	.717(.002)	.713(.003)	.711(.004)	.718(.002)

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Table 4 | The experimental results on an additional ten datasets of real-world single-cell for classification task, with performance evaluated by accuracy

Adam							**Alemany-Fin**						
Data	**Origin**	**MAMix**	**iSMOTE**	**SCVI**	**cscGAN**	**EDG**	**Data**	**Origin**	**MAMix**	**iSMOTE**	**SCVI**	**cscGAN**	**EDG**
n = 1	.198(.036)	–	–	–	–	**.427(.057)**	*n*=1	.297(.016)	–	–	–	–	**.522(.081)**
n = 2	.263(.066)	–	–	–	–	**.659(.068)**	*n*=2	.356(.078)	–	–	–	–	**.714(.054)**
n = 5	.513(.079)	.605(.026)	.580(.024)	–	–	**.807(.038)**	*n*=5	.467(.117)	.650(.035)	.584(.027)	–	–	**.877(.018)**
n = 10	.688(.057)	.759(.024)	.742(.025)	–	–	**.830(.022)**	*n*=10	.771(.065)	.837(.025)	.816(.007)	–	–	**.916(.006)**
n = 20	.824(.049)	.845(.020)	.836(.022)	.856(.019)	.846(.018)	**.877(.016)**	*n*=20	.868(.021)	.885(.004)	.876(.004)	.896(.003)	.870(.005)	**.923(.005)**
Bach							Guo						
Data	Origin	MAMix	iSMOTE	SCVI	cscGAN	EDG	Data	Origin	MAMix	iSMOTE	SCVI	cscGAN	EDG
n = 1	.335(.103)	–	–	–	–	**.447(.153)**	*n* = 1	.493(.103)	–	–	–	–	**.605(.074)**
n = 2	.432(.136)	–	–	–	–	**.726(.112)**	*n* = 2	.692(.094)	–	–	–	–	**.760(.133)**
n = 5	.789(.080)	.844(.025)	.832(.018)	–	–	**.899(.032)**	*n* = 5	.859(.033)	.872(.026)	.869(.018)	–	–	**.900(.020)**
n = 10	.863(.054)	.890(.019)	.872(.023)	–	–	**.926(.003)**	*n* = 10	.894(.026)	.916(.020)	.907(.006)	–	–	**.931(.004)**
n = 20	.880(.056)	.905(.006)	.894(.007)	.902(.007)	.895(.006)	**.934(.004)**	*n* = 20	.901(.036)	.924(.021)	.913(.013)	.927(.012)	.919(.013)	**.942(.006)**
Karaiskos							Li-Mammary						
Data	Origin	MAMix	iSMOTE	SCVI	cscGAN	EDG	Data	Origin	MAMix	iSMOTE	SCVI	cscGAN	EDG
n = 1	.516(.388)	–	–	–	–	**.754(.186)**	*n* = 1	.446(.130)	–	–	–	–	**.698(.099)**
n = 2	.662(.335)	–	–	–	–	**.841(.092)**	*n* = 2	.573(.164)	–	–	–	–	**.836(.034)**
n = 5	.877(.035)	.892(.026)	.887(.019)	–	–	**.918(.057)**	*n* = 5	.838(.038)	.852(.039)	.848(.043)	–	–	**.879(.044)**
n = 10	.908(.045)	.948(.051)	.938(.021)	–	–	**.977(.007)**	*n* = 10	.860(.041)	.886(.018)	.891(.021)	–	–	**.929(.004)**
n = 20	.970(.013)	.976(.007)	.977(.005)	.979(.004)	.968(.022)	**.980(.004)**	*n* = 20	.924(.040)	.929(.013)	.927(.005)	.936(.002)	.930(.003)	**.939(.007)**
MacParland							Muraro						
Data	Origin	MAMix	iSMOTE	SCVI	cscGAN	EDG	Data	Origin	MAMix	iSMOTE	SCVI	cscGAN	EDG
n = 1	.352(.171)	–	–	–	–	**.523(.263)**	*n* = 1	.477(.118)	–	–	–	–	**.544(.170)**
n = 2	.688(.042)	–	–	–	–	**.754(.078)**	*n* = 2	.546(.085)	–	–	–	–	**.809(.050)**
n = 5	.812(.041)	.829(.044)	.815(.059)	–	–	**.856(.024)**	*n* = 5	.830(.084)	.863(.037)	.848(.043)	–	–	**.937(.011)**
n = 10	.826(.051)	.867(.025)	.843(.028)	–	–	**.917(.013)**	*n* = 10	.913(.034)	.929(.038)	.920(.010)	–	–	**.958(.007)**
n = 20	.869(.038)	.893(.004)	.883(.003)	.907(.002)	.901(.003)	**.943(.009)**	*n* = 20	.953(.007)	.957(.005)	.954(.006)	**.959(.005)**	.956(.006)	.955(.003)
Oihane							Zilionis						
Data	Origin	MAMix	iSMOTE	SCVI	cscGAN	EDG	Data	Origin	MAMix	iSMOTE	SCVI	cscGAN	EDG
n = 1	.267(.054)	–	–	–	–	**.612(.096)**	*n* = 1	.431(.340)	–	–	–	–	**.747(.111)**
n = 2	.431(.098)	–	–	–	–	**.740(.048)**	*n* = 2	.457(.328)	–	–	–	–	**.808(.084)**
n = 5	.698(.045)	.751(.011)	.730(.045)	–	–	**.856(.006)**	*n* = 5	.643(.307)	.855(.064)	.742(.042)	–	–	**.939(.040)**
n = 10	.814(.035)	.850(.017)	.839(.015)	–	–	**.895(.014)**	*n* = 10	.883(.051)	.912(.045)	.925(.012)	–	–	**.974(.010)**
n = 20	.874(.013)	.905(.010)	.892(.010)	.882(.011)	.901(.010)	**.916(.005)**	*n* = 20	.963(.004)	.974(.002)	.975(.002)	.984(.002)	.980(.001)	**.986(.003)**

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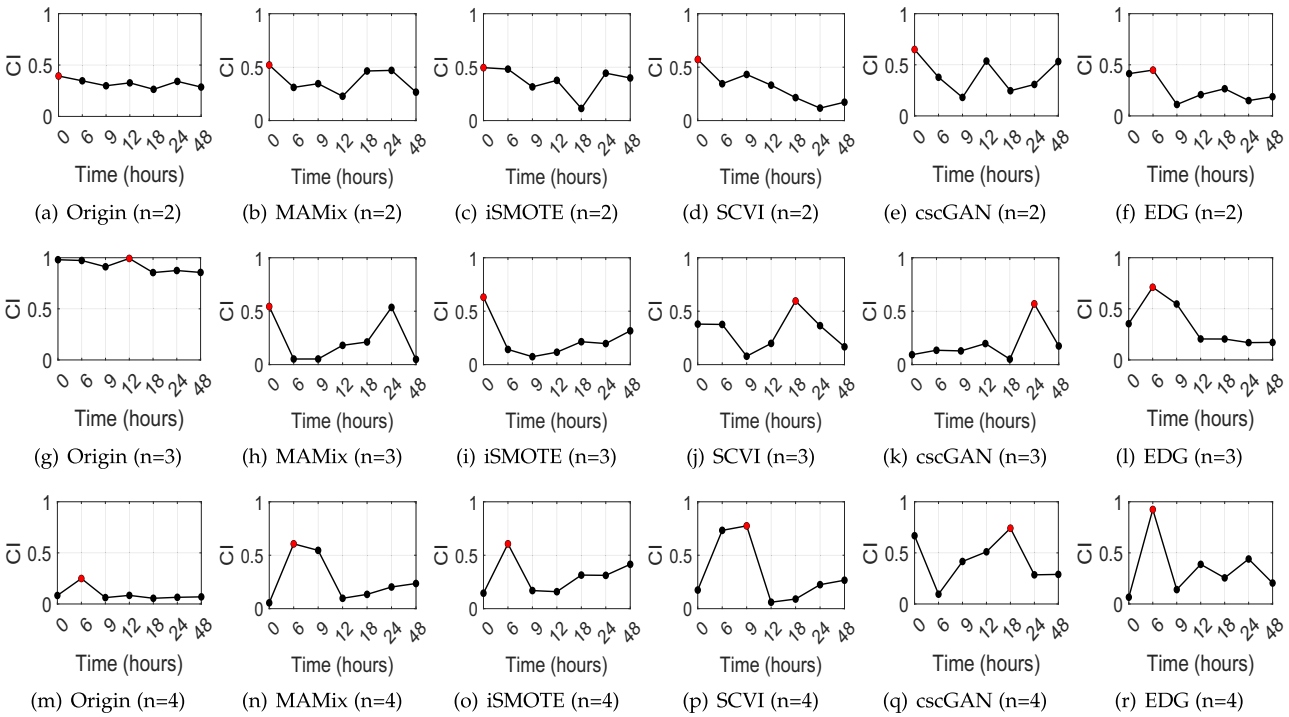


Fig. 3 | Tipping point prediction on a real-world dataset by employing dynamical network biomarkers (DNB) applied to both the original n data and the data generated by MAMix, iSMOTE, SCVI, cscGAN and the proposed EDG. The red dots indicate predicted tipping points, with the second time point (6 h) being the ground truth tipping point. **a** Appiling DNB on the original data with $n = 2$ samples per time point. Applying DNB on datasets augmented with 5 additional samples per time point generated by MAMix (**b**), iSMOTE (**c**), SCVI (**d**), cscGAN (**e**), and EDG (**f**), respectively, in combination with the original data. Similarly, (**g**–**l**) corresponds to the results obtained when using the original dataset with $n = 3$ samples per time point (**g**) and the augmented datasets generated by MAMix (**h**), iSMOTE (**i**), SCVI (**j**), cscGAN (**k**) and EDG (**l**). Furthermore, (**m**–**r**) show the results for the original dataset with $n = 4$ samples per time point (**m**) and the augmented datasets generated by MAMix (**n**), iSMOTE (**o**), SCVI (**p**), cscGAN (**q**) and EDG (**r**). Source data is provided in "Source Data.xlsx".

in combination with the original data. Similarly, Fig. 3g–l corresponds to the results obtained using the original dataset with 3 samples per time point (Fig. 3g) and the augmented datasets generated by MAMix (Fig. 3h), iSMOTE (Fig. 3i), SCVI (Fig. 3j), cscGAN (Fig. 3k) and EDG (Fig. 3l). The results demonstrate that DNB fails to predict the correct tipping point using only 2 or 3 samples, even when augmented with data generated by MAMix, iSMOTE, SCVI or cscGAN. The primary reason is that the limited sample size results in minor variance differences across time points, leading to inaccurate predictions. Furthermore, MAMix, iSMOTE, SCVI and cscGAN require certain samples for sample combination and model training, failing to generate effective samples in scenarios with scarce sample size and high-dimensional features. In contrast, Fig. 3m–r indicates employing DNB on the original dataset with 4 samples per time point (Fig. 3m) and augmented dataset generated by MAMix (Fig. 3n), iSMOTE (Fig. 3o), SCVI (Fig. 3p), cscGAN (Fig. 3q) and EDG (Fig. 3r). Moreover, using the samples generated by EDG further improves prediction accuracy, with CI score increasing more sharply and consistently, providing more precise, effective, and robust early-warning signals of the critical transition (Fig. 3r), compared with the original one (Fig. 3m). This improvement is attributed to EDG with generating high-quality samples with aligned features. EDG magnifies variance differences across time points, enabling more accurate identification of critical states or tipping points.

Furthermore, we evaluate the effectiveness of EDG by visualizing the generated samples on a human face dataset borrowed from ref. 45, as shown in Fig. 4. "Origin" refers to original data, consisting of 5 samples, while "Generation" represents 10 newly generated samples by EDG. As illustrated in Fig. 4a and Fig. 4b, the generated samples maintain meaningful representations of original data while exhibiting significant diversity. This demonstrates that EDG can effectively capture underlying distribution and generate samples that align well with the features of the original data. In addition, EDG offers flexibility in generating samples with varying styles by adjusting the model's hyperparameters. This capability is illustrated in Fig. 4c, d, where the generated samples exhibit distinct stylistic variations compared to the original data. Such transformations demonstrate the adaptability of EDG in creating data that not only reflects the original distribution but also introduces meaningful variations. These results highlight the robustness and versatility of EDG in generating high-quality samples that exhibit both rich diversity and stylistic variability.

Finally, additional experiments on ablation study of OT and HSIC on single-cell classification, parameter sensitivity analysis, computational cost on original feature dimensionality and generated sample size, classification accuracy with respect to original / generated sample size, out-of-distribution experiment, experimental results compared with ChatGPT under extremely limited samples, and the selection of generation multiplier M are reported in Supplementary Table 6 in Supplementary Note 4, Supplementary Fig. 1 in Supplementary Note 5, Supplementary Fig. 2 in Supplementary Note 6, Supplementary Fig. 3 in Supplementary Note 7, Supplementary Table 7 in Supplementary Note 8, Supplementary Table 8 in Supplementary Note 9, and Supplementary Note 10, respectively.

In conclusion, the above experimental results demonstrate that the proposed EDG achieves competitive performance across multiple tasks, including classification, feature selection, clustering, and tipping point prediction. Specifically, EDG effectively assists in training classifiers, identifies features with enhanced discriminative ability, learns more robust and accurate clustering centroids, and improves the accuracy of tipping point prediction, especially under extreme data

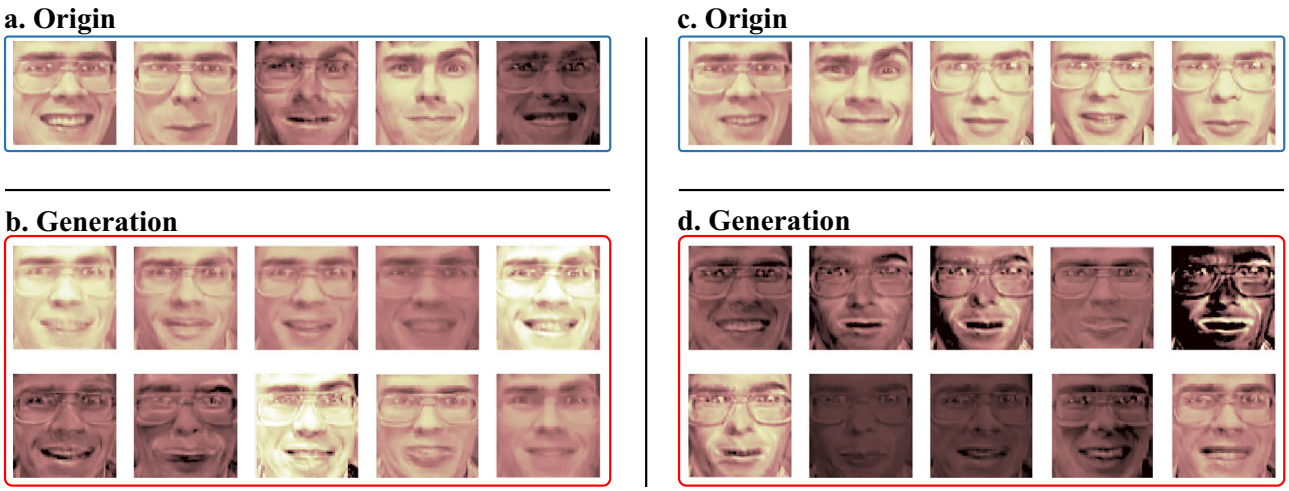


Fig. 4 | Visualization of samples generated by the proposed EDG. a, **c** The original data consisting of 10 samples. **b**, **d** The newly generated 20 samples by EDG with rich diversity.

scarcity. This demonstrates that EDG provides a reliable and effective solution for generating high-quality and informative single-cell data. The superior performance of EDG is attributed to its ability to exploit the inherent low-dimensional principles embedded in high-dimensional data to facilitate the generation of high-quality, aligned samples.

Discussion

In this study, we proposed an explainable generation framework under extreme sample scarcity by exploiting high-dimensional information rather than relying on large numbers of training samples, as in the traditional methods, named by Explainable Data Generation (EDG) based on matrix tri-factorization. The primary contributions are summarized as follows:

Designed goal on mitigating challenges of biomedical scenarios under extreme sample scarcity: The proposed EDG is designed for real-world biomedical scenarios, especially on extreme cases such as early-stage or emerging biomedical conditions (e.g., Patient Zero of a brand-new disease with only one sample). It aims to address the challenges of data scarcity and high-dimensional feature spaces across multiple critical tasks.

New data generation approach: This paper introduces an explainable data generation approach, named by EDG. It leverages the inherent low-dimensional principles embedded in observed high-dimensional data, which retains comprehensive information for new data generation under extremely limited data regimes while maintaining interpretability and data structure.

Explainable data generation approach: The explainable nature of EDG is a significant advantage over black-box deep-learning approaches. The use of matrix tri-factorization provides a clear and interpretable decomposition of data into latent representations, enhancing the transparency of the data generation process. Moreover, the generated samples serve as feature bases for the original samples, enabling the generated data to be both explainable and biologically meaningful. These characteristics not only improves trust in the generated data but also enable researchers to gain deeper insights into the underlying structure of the data, enabling the generated data explainable and faithful preservation of biological characteristics.

Low-dimensional representation with larger sample size: In contrast to traditional methods that employ low-dimensional latent representations with an equal sample size as the observed samples, EDG represents small high-dimensional samples using low-dimensional latent representations with a larger sample size via matrix tri-factorization with HSIC and OT, thus leading to new sample generation and also mitigating the challenges of scarce data and high-dimensional features.

Controlled extension via regularization: In EDG, matrix tri-factorization cooperating with HSIC and OT is utilized to guarantee the generation process stable with the explicit guidance of keeping consistent with the global structure and distribution of the original data. Although the base factorization is linear, the incorporation of OT and HSIC regularizations allows EDG to capture certain nonlinear dependencies and distributional alignment in the data. These terms effectively extend the representational flexibility of the framework while maintaining interpretability.

Despite its strong performance, the proposed frameworks has certain limitations that warrant further investigation. The primary limitations and future work are shown as follows:

Not applicable scenarios: The EDG framework is primarily designed for extremely few-shot high-dimensional biomedical datasets, which is motivated by leveraging the inherent low-dimensional principles embedded within the observed high-dimensional data. Consequently, for large-scale, low-dimensional, or unstructured datasets (e.g., natural images in CV or text in NLP), the deep generative models are more suitable with large-scale samples for pre-training. Furthermore, in the linear scenarios with extremely simple structures, the regularizations of OT and HSIC may introduce unnecessary computational cost and over-smoothing, reducing the natural diversity of the generated data.

Insufficient biological explainability: EDG mainly provides a more transparent generation process in comparison of fully black-box generative models, with structural interpretability at the model level. Although the learned latent bases of EDG may capture important data structures and feature dependencies, the current study does not establish a direct biological correspondence between these latent components and specific biological feature modules, signaling pathways, chromatin accessibility programs, or regulatory mechanisms. Consequently, future work will focus on integrating external biological knowledge, such as pathway annotations, gene regulatory networks, and functional enrichment analysis, to examine whether the learned latent bases can be associated with biologically meaningful modules or regulatory programs. We believe that such investigations would further strengthen the biological interpretability and practical utility of the proposed framework in biomedical applications.

Diverse extensions: The deep generative models have built-in diversity mechanisms, which are a helpful and promising direction for generation. In the future work, EDG can be integrated with diversity-promoting mechanisms (e.g., entropy maximization, variational

regularization, or adversarial objectives) to combine interpretability, biological meaning with creativity.

Dynamic extensions: While matrix tri-factorization is effective in capturing low-dimensional structures, its reliance on static data may limit its applicability in dynamic or time-evolving datasets, where temporal dependencies play a significant role. Future extensions could incorporate time-dependent modeling techniques, such as NeuralODE (Neural Ordinary Differential Equations), to better handle temporal dynamics, with aims to reconstruct the inherent principle of time-dependent data, even in cases where the data is unordered or with vast states missing.

Complementary perspective: The deep-learning-based methods are often less interpretable, resource-intensive, and prone to mode collapse in extreme few-shot settings, such as single-sample scenarios. One can employ the proposed EDG to generate new samples based on the scarce data, which could be further utilized for the pre-training of deep-learning-based methods because of their requirement of substantial amounts of training data for effective pre-training.

Methods

EDG Framework

Mathematically, consider an observed data matrix $\boldsymbol{X} \in \mathbb{R}^{d\times n}$ consisting of n samples, each characterized by d-dimensional features. Unlike traditional methods using low-dimensional latent representation with an equal sample size as the observed samples, the proposed EDG aims to represent the observed small (n) high-dimensional (d) samples X by low-dimensional (d') latent representation $\boldsymbol{Y}$ with a larger sample size (m) via matrix tri-factorization, thus leading to new sample generation and also mitigating the challenges of scarce data and high-dimensional features. In detail, as shown in eqn. (2), EDG decomposes the observed high-dimensional samples $\boldsymbol{X}$ to their intrinsic feature bases $\boldsymbol{Y}$, corresponding to a low-dimensional latent system, in not an equal sample size but a larger sample size basis. Thus, such generated feature bases $\boldsymbol{Y}$ not only reconstruct the original samples $\boldsymbol{X}$ (as shown in Fig. 1c) with Hilbert-Schmidt independence criterion (HSIC)[14,15], but also generate large numbers of new samples $\boldsymbol{X}^{u}$ (as shown in Fig. 1d) with optimal transport (OT)[16,17], with rich diversity and aligned features. This process ensures explainability through the linear combination of sample reconstruction, further enhancing the interpretability of the generated data. The process of matrix tri-factorization for new sample generation is defined as

$$\begin{aligned}&\min_{\boldsymbol{U},\boldsymbol{Y},\boldsymbol{V}} ||\boldsymbol{X}-\boldsymbol{U}\boldsymbol{Y}\boldsymbol{V}||_F^2\\ &s.t.\boldsymbol{U}\mathbf{1}=\mathbf{1},\boldsymbol{V}^T\mathbf{1}=\mathbf{1}\end{aligned} \tag{5}$$

where $\mathbf{1}$ denotes a column vector with all elements equal to 1 which are utilized to enforce the constraints that the column sums of $\boldsymbol{U}$ and the row sums of $\boldsymbol{V}$ equal to 1. In detail, $\boldsymbol{U} \in \mathbb{R}^{d\times d'}$ serves as a projection matrix that reduces the feature dimensionality from d-dimensions to d'-dimensions, with $d' < d$. The matrix $\boldsymbol{Y} \in \mathbb{R}^{d'\times m}$ represents the generated feature basis consisting of m samples with d'-dimensional features, with $m > n$. The matrix product of $\boldsymbol{Y}^{\boldsymbol{v}} = \boldsymbol{Y}\boldsymbol{V} \in \mathbb{R}^{d'\times n}$ is the low-dimensional representation of the observed data $\boldsymbol{X} \in \mathbb{R}^{d\times n}$, which is also traditional dimension reduction. Consequently, Y is the feature basis of $\boldsymbol{Y}^{\boldsymbol{v}}$ and $\boldsymbol{V} \in \mathbb{R}^{m\times n}$ is the corresponding coefficient matrix. In this way, each sample in $\boldsymbol{Y}^{\boldsymbol{v}}$ can be reconstructed as a linear combination of the generated feature basis $\boldsymbol{Y}$ and the corresponding coefficient matrix $\boldsymbol{V}$. Furthermore, to ensure that the low-dimensional representation $\boldsymbol{Y}^{\boldsymbol{v}}$ accurately captures the structure of the observed data, HSIC is employed to maximize the distributional consistency between the generated low-dimensional representation $\boldsymbol{Y}^{\boldsymbol{v}} = \boldsymbol{Y}\boldsymbol{V}$ and original data $\boldsymbol{X}$. This allows $\boldsymbol{Y}^{\boldsymbol{v}}$ to effectively represent $\boldsymbol{X}$ in the low-dimensional space, with $\boldsymbol{Y}$ serving as the feature basis of the original data. Finally, to generate samples in the original feature space with aligned features, the generated feature basis $\boldsymbol{Y}$ is mapped back to the original feature space of $\boldsymbol{X}$ cooperating with the project matrix $\boldsymbol{U}$. The final generated samples with aligned features is denoted as $\boldsymbol{X}^{\boldsymbol{u}} = \boldsymbol{U}\boldsymbol{Y} \in \mathbb{R}^{d\times m}$, with $m > n$. In addition, OT is employed to ensure the alignment and consistency between the distributions of the original high-dimensional data $\boldsymbol{X}$ and the generated data $\boldsymbol{X}^{\boldsymbol{u}}$ so that the generated samples can accurately reflect the original data's structure and distribution, with reinforcing interpretability and reliability of the proposed EDG.

Finally, the complete loss function of the proposed EDG framework is formulated as follows:

$$\begin{aligned}&\min_{\boldsymbol{U},\boldsymbol{Y},\boldsymbol{V}} \lambda_M||\boldsymbol{X}-\boldsymbol{U}\boldsymbol{Y}\boldsymbol{V}||_F^2\\ &+\lambda_H HSIC(\boldsymbol{X},\boldsymbol{Y}\boldsymbol{V})+\lambda_O OT(\boldsymbol{X},\boldsymbol{U}\boldsymbol{Y})\\ &s.t.\boldsymbol{U}\mathbf{1}=\mathbf{1},\boldsymbol{V}^T\mathbf{1}=\mathbf{1}\end{aligned} \tag{6}$$

where λ_M, λ_H, and λ_O are hyperparameters to balance different terms.

To obtain the numerical solution for the proposed EDG, we employ the widely used alternating optimization scheme. This iterative approach updates each variable sequentially while keeping the others fixed, thereby breaking down the complex optimization problem into a series of simpler subproblems. Specifically, at each iteration, one variable is optimized by solving the corresponding subproblem, while the remaining variables are treated as constants. This process is repeated iteratively until convergence is achieved, ensuring that the overall objective function is minimized effectively. In detail, the optimization stopping criteria are defined as a maximum number of iterations: 10 (default setting across datasets). In practice, we observe stable convergence well before reaching the maximum iteration limit for most datasets. Furthermore, the hyperparameters are set as $\lambda_M = 0.8$, $\lambda_H = 0.2$, and $\lambda_O = 0.2$ across datasets.

Furthermore, in EDG, matrix tri-factorization cooperating with HSIC and OT is utilized to guarantee the generation process stable with the explicit guidance of keeping consistent with the global structure and distribution of the original data for downstream predictive tasks. $HSIC(\boldsymbol{X},\boldsymbol{Y}\boldsymbol{V})$ indicates employing Hilbert-Schmidt independence criterion (HSIC)[14,15] to maximize the distributional consistency between the generated low-dimensional representation $\boldsymbol{Y}^{\boldsymbol{v}} = \boldsymbol{Y}\boldsymbol{V} \in \mathbb{R}^{d'\times n}$ and the original data $\boldsymbol{X} \in \mathbb{R}^{d\times n}$, to ensure that the low-dimensional representation $\boldsymbol{Y}^{\boldsymbol{v}}$ accurately captures the structure of the observed data. Here, we provide a brief overview of the widely recognized formulation of HSIC.

Mathematically, HSIC measures the dependence between two observations, $\mathbf{S}^x$ and $\mathbf{S}^y$, which correspond to two different feature spaces, each consisting of n samples. Specifically, $\{(\mathbf{s}_i^x,\mathbf{s}_i^y) \in \mathcal{X}\times\mathcal{Y}\}_{i=1}^n$ are drawn jointly from a probability distribution $P_{\mathbf{S}^x\mathbf{S}^y}$, where the dependence between $\mathbf{s}^x$ and $\mathbf{s}^y$ is quantified by HSIC. The dependence is computed based on the norm of the cross-covariance operator over the domain $\mathcal{X}\times\mathcal{Y}$ in Hilbert space. A higher HSIC value indicates stronger dependence, determined by the choice of kernel functions. HSIC is formally defined as $HSIC(P_{\mathbf{S}^x\mathbf{S}^y},\mathcal{F},\mathcal{G}) := \left\|C_{\mathbf{S}^x\mathbf{S}^y}\right\|_{\mathrm{HS}}^2$, where $\|\mathbf{A}\|_{\mathrm{HS}} = \sqrt{\sum_{i,j}a_{ij}^2}$ represents the Hilbert-Schmidt norm. $\mathcal{F}$ and $\mathcal{G}$ are the reproducing kernel Hilbert Space (RKHS) corresponding to $\mathcal{X}$ and $\mathcal{Y}$, respectively. The cross-covariance operator, $C_{\mathbf{S}^x\mathbf{S}^y}$, captures the covariance between two random variables and is defined as $C_{\mathbf{S}^x\mathbf{S}^y} = \mathbb{E}_{\mathbf{S}^x\mathbf{S}^y}\left[\left(\boldsymbol{\phi}(\mathbf{s}^x)-\mu_{\mathbf{s}^x}\right)\otimes\left(\boldsymbol{\varphi}(\mathbf{s}^y)-\mu_{\mathbf{s}^y}\right)\right]$, where $\mu_{\mathbf{s}^x} = \mathbb{E}(\boldsymbol{\phi}(\mathbf{s}^x)), \mu_{\mathbf{s}^y} = \mathbb{E}(\boldsymbol{\varphi}(\mathbf{s}^y))$, and $\otimes$ is denotes the tensor product. The mapping $\boldsymbol{\phi}(\mathbf{s}^x)$ and $\boldsymbol{\varphi}(\mathbf{s}^y)$ transform $\mathbf{s}^x \in \mathcal{X}$ and $\mathbf{s}^y \in \mathcal{Y}$ into kernel space $\mathcal{F}$ and $\mathcal{G}$, respectively, with kernel functions $k_x\left(\mathbf{s}_i^x,\mathbf{s}_j^x\right) = <\boldsymbol{\phi}\left(\mathbf{s}_i^x\right),\boldsymbol{\phi}\left(\mathbf{s}_j^x\right)>$ and $k_y\left(\mathbf{s}_i^y,\mathbf{s}_j^y\right) = <\boldsymbol{\varphi}\left(\mathbf{s}_i^y\right),\boldsymbol{\varphi}\left(\mathbf{s}_j^y\right)>$. Accordingly, the empirical HSIC is defined as:

$$HSIC(\mathbf{S}^x,\mathbf{S}^y) = (n-1)^{-2}\mathrm{tr}\left(\mathbf{K}_x\mathbf{H}\mathbf{K}_y\mathbf{H}\right) \tag{7}$$

where $\mathbf{K}_x$ and $\mathbf{K}_y$ are the Gram matrices with $k_{x,ij} = k_x\left(\mathbf{s}_i^x, \mathbf{s}_j^x\right), k_{y,ij} = k_y\left(\mathbf{s}_i^y, \mathbf{s}_j^y\right)$. The centering matrix $\boldsymbol{H}$ is defined as 7D1$h_{ij} = \delta_{ij} - 1/n$, which ensures that the Gram matrices have zero mean in the feature space.

In our implementation, we utilize the inner product as the kernel function. Specifically, the Gram matrices are computed as $\mathbf{K}^x = (\mathbf{S}^x)^T\mathbf{S}^x = \mathbf{X}^T\mathbf{X}$, and $\mathbf{K}^y = (\mathbf{S}^y)^T\mathbf{S}^y = \mathbf{V}^T\mathbf{Y}^T\mathbf{Y}\mathbf{V}$. Maximizing $HSIC(\mathbf{S}^x, \mathbf{S}^y)$ increases the dependency between $\mathbf{K}^x$ and $\mathbf{K}^y$, effectively penalizing any disagreement between the kernel matrices from different feature spaces.

The overall formula of $HSIC(\boldsymbol{X}, \boldsymbol{YV})$ is presented as follows:

$$\begin{aligned} &\min_{\boldsymbol{Y},\boldsymbol{V}} HSIC(\boldsymbol{X}, \boldsymbol{YV}) \\ &= \min_{\boldsymbol{Y},\boldsymbol{V}} -(n-1)^{-2}\mathrm{tr}\left(\mathbf{K}_x\mathbf{H}\mathbf{K}_y\mathbf{H}\right) \\ &= \min_{\boldsymbol{Y},\boldsymbol{V}} -(n-1)^{-2}\mathrm{tr}\left(\mathbf{X}^T\mathbf{X}\mathbf{H}\mathbf{V}^T\mathbf{Y}^T\mathbf{Y}\mathbf{V}\mathbf{H}\right) \\ &s.t.\boldsymbol{V}^T\mathbf{1} = \mathbf{1} \end{aligned} \tag{8}$$

Furthermore, $OT(\boldsymbol{X}, \boldsymbol{UY})$ indicates utilizing optimal transport (OT)[16,17] to ensure the alignment and consistency between the distributions of the original high-dimensional data $\boldsymbol{X} \in \mathbb{R}^{d\times n}$ and the generated data $\boldsymbol{X^u} = \boldsymbol{UY} \in \mathbb{R}^{d\times m}$, so that the generated samples can accurately reflect the original data's structure and distribution, with reinforcing interpretability and reliability of the proposed EDG. Here, we provide a concise overview of the widely recognized formulation of OT.

Mathematically, given the cost matrix $\boldsymbol{C} \in {}^{n\times m}$, the objective of OT is to determine the optimal coupling matrix (also referred to as the transport plan) $\boldsymbol{T}^* \in {}^{n\times m}$. The optimization problem is formulated as follows:

$$\min_{\boldsymbol{T}\in D(a,b)} <\boldsymbol{C}, \boldsymbol{T}> \tag{9}$$

where the computation of $<C, T> = \sum_{i,j} c_{ij}t_{ij}$. The transport polytope of a and b is defined as $D(a,b) = \{\mathbf{T} \in \mathbb{R}^{n\times m} | \mathbf{T}\mathbf{1}_m = a, \mathbf{T}^\top\mathbf{1}_n = b\}$, which can also be interpreted as a set of all possible joint probabilities of two distributions. a and b belongs to the probability simplex $\Sigma_n := \{a \in \mathbb{R}^n | a^\top\mathbf{1}_n = 1\}$, where $\mathbf{1}_d$ is a d-dimensional vector of ones. Typically, a and b are initialized as uniform distribution, given by $a = \frac{1}{n}1_n$ and $b = \frac{1}{m}1_m$.

Subsequently, a entropic regularization term $\mathbf{H}(\boldsymbol{T})$ is introduced into the optimization problem, as described below:

$$\min_{T\in D(a,b)} <C, T> - \varepsilon\mathbf{H}(\boldsymbol{T}) \tag{10}$$

where $\mathbf{H}(\boldsymbol{T}) = -\sum_{i,j} t_{ij}\log\left(t_{ij}\right)$ and ε is balanced parameter.

However, the standard OT formulation is not well-suited for scenarios involving partial transport or displacement problems, where strict marginal constraints are impractical or undesirable. To address this limitation, unbalanced optimal transport (UOT)[46,47] has been proposed, which introduces a relaxation of the marginal constraints through soft penalties based on the Kullback-Leibler (KL) divergence. The UOT problem is formulated as follows:

$$\begin{aligned} &\min_{\mathbf{T}\in\mathbb{R}^{n\times m}} <\mathbf{C}, \mathbf{T}> - \varepsilon H(\mathbf{T}) \\ &+\rho\left(D_{KL}\left(\mathbf{T}\mathbf{1}_m \parallel \mathbf{a}\right) + \rho D_{KL}\left(\mathbf{T}^\top\mathbf{1}_n \parallel \mathbf{b}\right)\right) \end{aligned} \tag{11}$$

where D_{KL} denotes Kullback-Leibler Divergence, and ρ represents the balanced parameter. This formulation enables partial mass transport, making it particularly suitable for applications where strict marginal conservation is infeasible or undesirable.

The resulting optimization problem is strictly convex and can be efficiently solved using the inexact proximal point method (IPOT)[48], formulated as follows:

$$\boldsymbol{\alpha}^{(l+1)} = \left(\frac{\mathbf{a}}{\mathbf{G}\boldsymbol{\beta}^{(l)}}\right)^{\frac{\rho}{\rho+\varepsilon}}, \boldsymbol{\beta}^{(l+1)} = \left(\frac{\mathbf{b}}{\mathbf{G}^\top\boldsymbol{\alpha}^{(l)}}\right)^{\frac{\rho}{\rho+\varepsilon}} \tag{12}$$

initializing from $\boldsymbol{\beta}^{(0)} = \frac{1}{n}\mathbf{1}_n$, where $\mathbf{g}_{ij} = \mathbf{t}_{ij}^{(l)}e^{-\mathbf{c}_{ij}/\epsilon}$. The optimal coupling matrix (transport plan) $\mathbf{T}^*$ with $\mathbf{t}_{ij}^* = \alpha_i\mathbf{g}_{ij}\beta_j$.

After obtaining the optimal transport plan $\mathbf{T}^*$, the cost matrix $C \in {}^{n\times m}$ is updated to reflect the transport costs between the original high-dimensional data $\boldsymbol{X} \in \mathbb{R}^{d\times n}$ and the generated data $\boldsymbol{X^u} = \boldsymbol{UY} \in \mathbb{R}^{d\times m}$. In practice, the Euclidean distance is commonly used as the transport cost, defined as follows:

$$c_{ij} = ||x_i - x_j^u||^2 \tag{13}$$

In addition, we offer an alternative approach for calculating the transport cost. Inspired by refs. 49,50, the correlation matrix can be employed as a substitute for the traditional distance matrix. Using this formulation, the optimal coupling matrix (transport plan) $\mathbf{T}^*$ is determined by solving the following optimization problem:

$$\begin{aligned} &\min_{\mathbf{T}\in\mathbb{R}^{n\times m}} tr(\boldsymbol{Y}^T\boldsymbol{U}^T\boldsymbol{X}\boldsymbol{T}) - \varepsilon H(\mathbf{T}) \\ &+\rho\left(D_{KL}\left(\mathbf{T}\mathbf{1}_m \parallel \mathbf{a}\right) + \rho D_{KL}\left(\mathbf{T}^\top\mathbf{1}_n \parallel \mathbf{b}\right)\right) \end{aligned} \tag{14}$$

The use of a correlation matrix can enhance the feature alignment between the original high-dimensional data $\boldsymbol{X}$ and the generated data $\boldsymbol{X^u} = \boldsymbol{UY}$ during the transport process. To solve the resulting optimization problem, the widely adopted alternating optimization scheme is employed. This challenging optimization task, involving two variables, the $\boldsymbol{C}$ and the coupling matrix $\boldsymbol{T}$ is decomposed into a series of simpler subproblems. At each iteration, one variable is optimized by solving its corresponding subproblem while keeping the other variable fixed. This sequential optimization process is repeated iteratively until convergence is achieved, ensuring effective minimization of the overall objective function.

Statistics & reproducibility

All statistical calculations were implemented in MATLAB (version R2023a, MathWorks). The detailed statistical analyses were indicated in tables and figures. Complete randomization was performed for allocating training and testing groups. No statistical method was used to predetermine sample size. No data were excluded from the analyses. Our study does not involve group allocation that requires blinding.

Implementation details

All experiments were conducted on a workstation equipped with an Intel(R) Core(TM) i9-12900K CPU (3.2 GHz), and 128 GB RAM memory.

The proposed EDG framework involves three key hyperparameters, namely the weights of the matrix factorization, HSIC regularization, and OT regularization, denoted as $\lambda_M, \lambda_H, \lambda_O$, respectively. In all experiments, we set $\lambda_M = 0.8$, $\lambda_H = 0.2$, and $\lambda_O = 0.2$, which were found to provide stable performance across datasets.

For the hyperparameter of multiplier M, relatively larger multipliers (e.g., $M \geq 10$) are recommended to sufficiently enrich the training distribution under extremely limited sample conditions ($n \leq 10$), while smaller multipliers (e.g., $M \leq 10$) are generally adequate and may provide better generalization performance when a moderate number of real samples is available ($n \geq 20$).

In our experiments, for the supervised tasks, including classification, feature selection, and tipping point prediction, the observed data belongs to each category or each time point, allowing the generated data to be annotated with the corresponding ground truth. For the

unsupervised clustering task, however, the observed data included samples from all categories. At last, generated synthetic samples are mixed with real samples using equal augmentation ratios across classes to preserve the original class distribution observed in real-world applications.

At last, the maximum number of iterations is set as 10 across datasets. In practice, we observe stable convergence well before reaching the maximum iteration limit for most datasets.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All datasets used in this study are already published and obtained from public data repositories[30,34] at https://cblast.gao-lab.org/download and https://scglue.readthedocs.io/en/latest/data.html. Any additional requests for information can be directed to, and will be fulfilled by, the corresponding author. Source data are provided in this paper.

Code availability

An open-source MATLAB (version R2023a, MathWorks) implementation of the proposed EDG package[51] (DOI: 10.5281/zenodo.21873700) is available at https://github.com/De-jun-Zhang/SmallSampleGeneration.

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Author contributions

L.C. supervised the project. H.C. and P.L. designed the experiments. B.Z. and P.T. carried out the experiments and wrote the manuscript. J.R., J.W., Q.W., S.C. and W.H. discussed the experimental results. All authors discussed the results and assisted during manuscript preparation.

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Competing interests

The authors declare no competing interests.

Additional information

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